Neural Population Dynamics Reveal Internal Representations Underlying Adaptive Obstacle Avoidance in Reinforcement Learning

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BACKGROUND & MOTIVATION

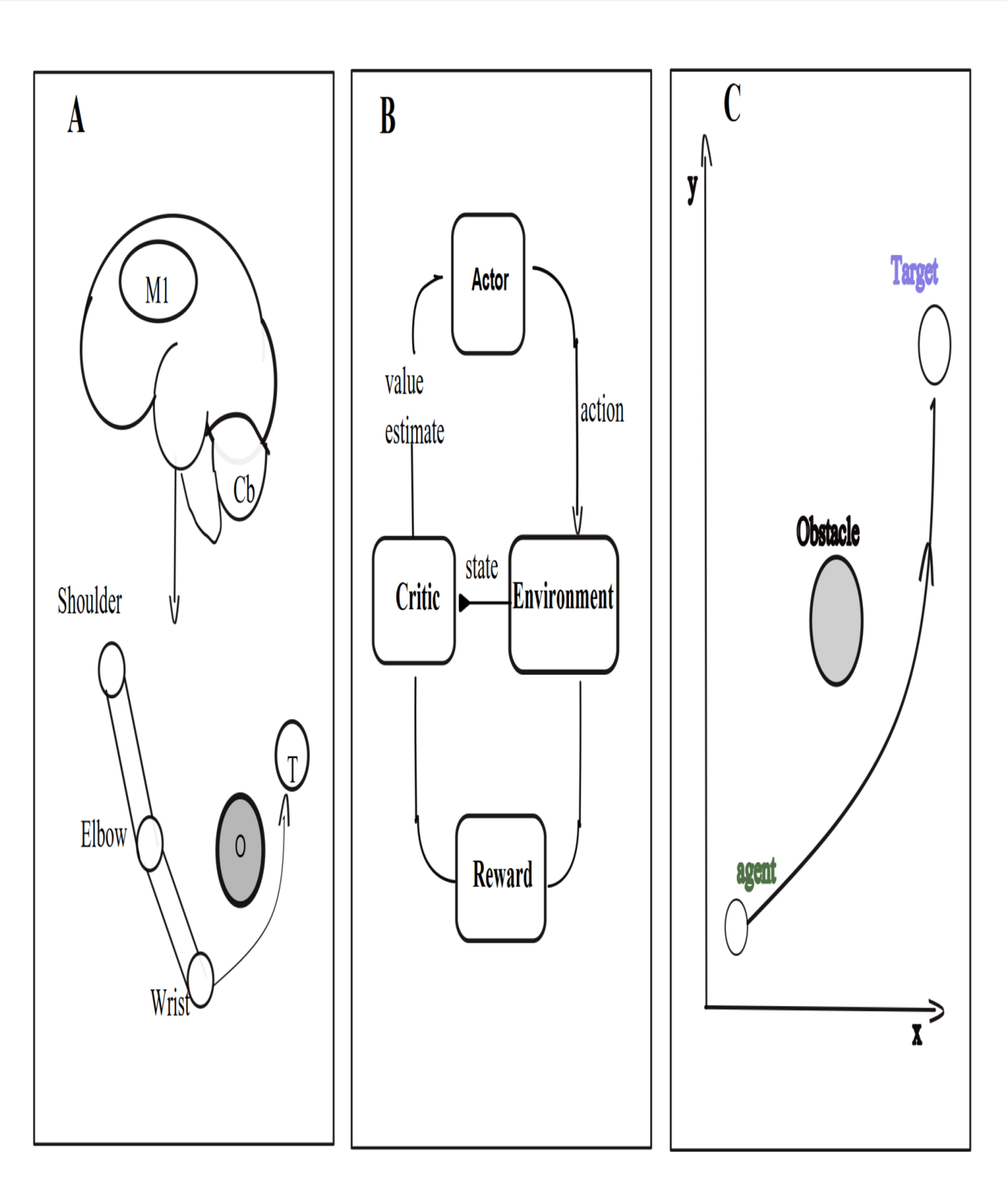
- Motor recovery after stroke is highly heterogeneous. Existing therapies lack mechanistic precision to tailor training to the individual patient.
- Reinforcement learning (RL) controllers trained by reward discover adaptive motor strategies analogously to biological motor learning.
- Motor cortex population activity evolves on compact low-dimensional manifolds. Whether RL controllers develop the same organisation is unknown.

We ask: does reward-driven learning produce structured neural population representations parallel to biological motor cortex?

METHODS

- 2-D point-mass agent navigates obstacle corridor to a fixed goal under Newtonian dynamics (Gymnasium environment).
- PPO actor-critic controller trained for 3×10^6 steps; two hidden layers × 256 units (feedforward MLP).
- 7 perturbation conditions: control (P0), leftward (L1–L3) and rightward (R1–R3) lateral force impulses.
- Population analysis: PCA · K-means · linear decoding applied to hidden-layer activations during evaluation.

Fig 1 · Task & Control Architecture



KEY FINDINGS

74.8 % variance in 2 PCs
Hidden-layer activity compresses into a compact low-dimensional manifold — mirroring motor cortex.

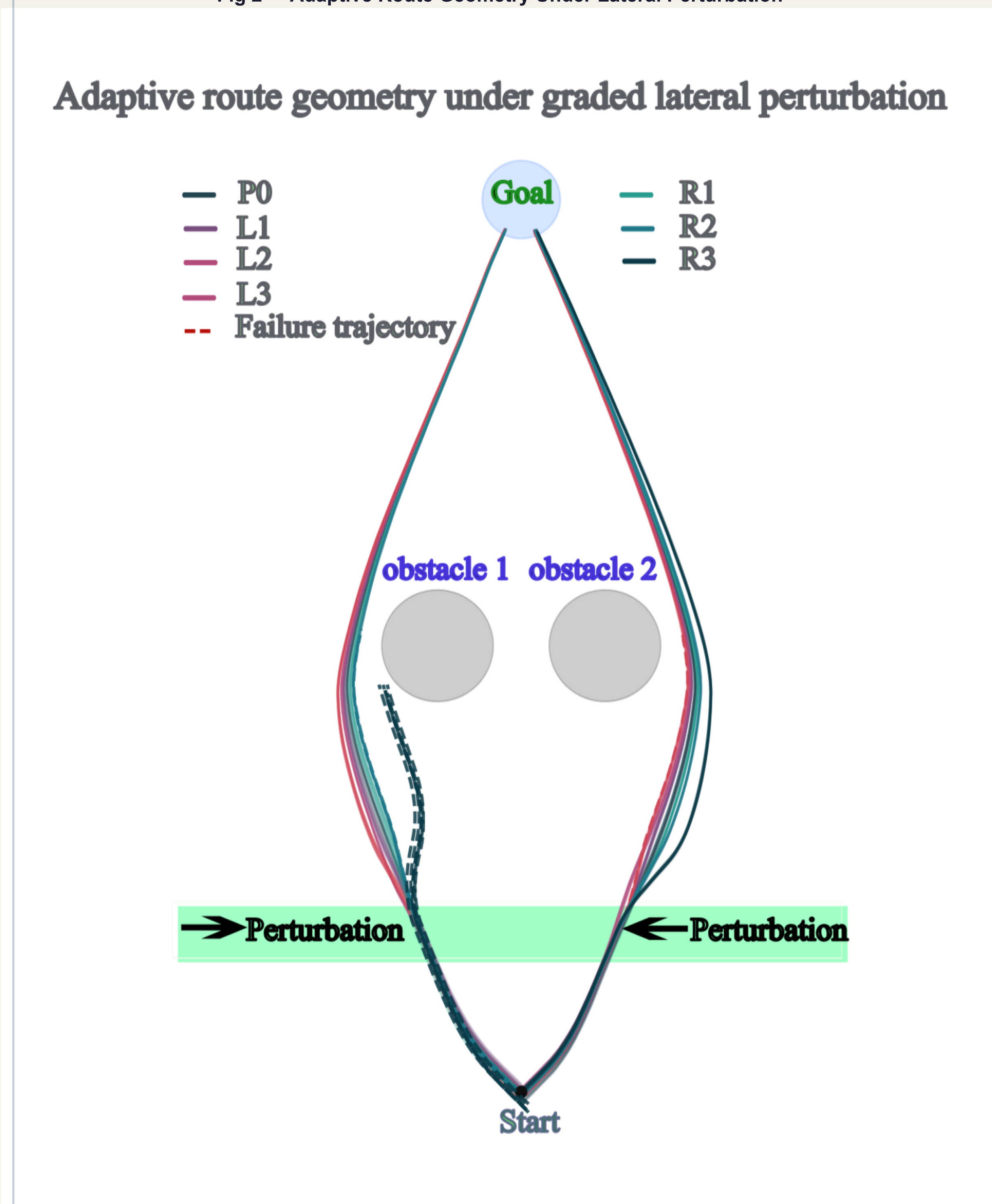
4 phase-aligned clusters
K-means reveals movement-phase structure (approach · perturbation · correction · arrival).

Direction asymmetry
Rightward perturbations are more disruptive due to learned left-biased corridor trajectory.

Linear decodability
Extreme conditions linearly separable; mild conditions geometrically indistinct — matching biological findings.

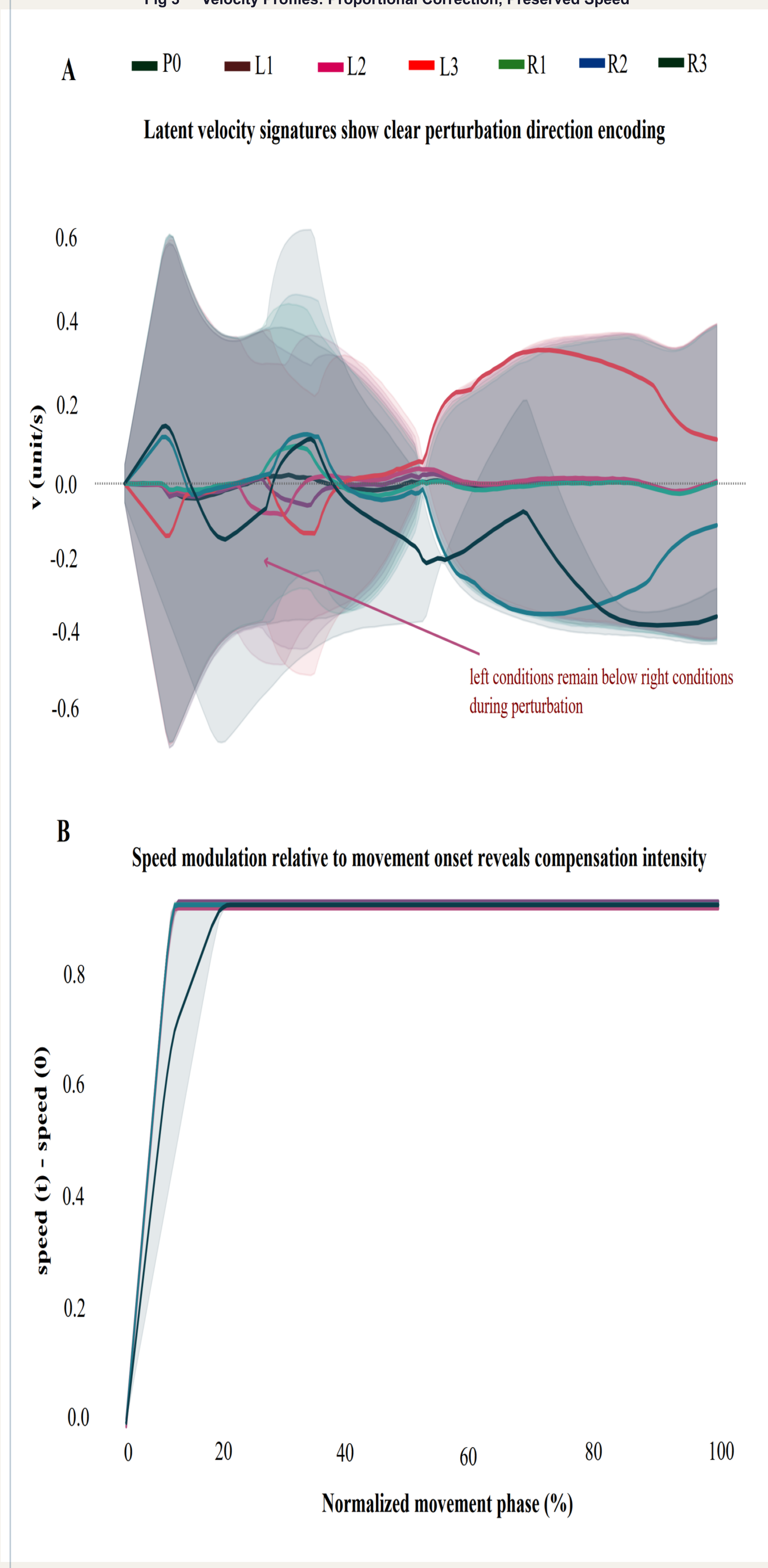
RESULTS

Fig 2 · Adaptive Route Geometry Under Lateral Perturbation



Solid = success · Dashed = failure · Shaded band = perturbation zone · Rightward conditions (R2, R3) show highest failure rate

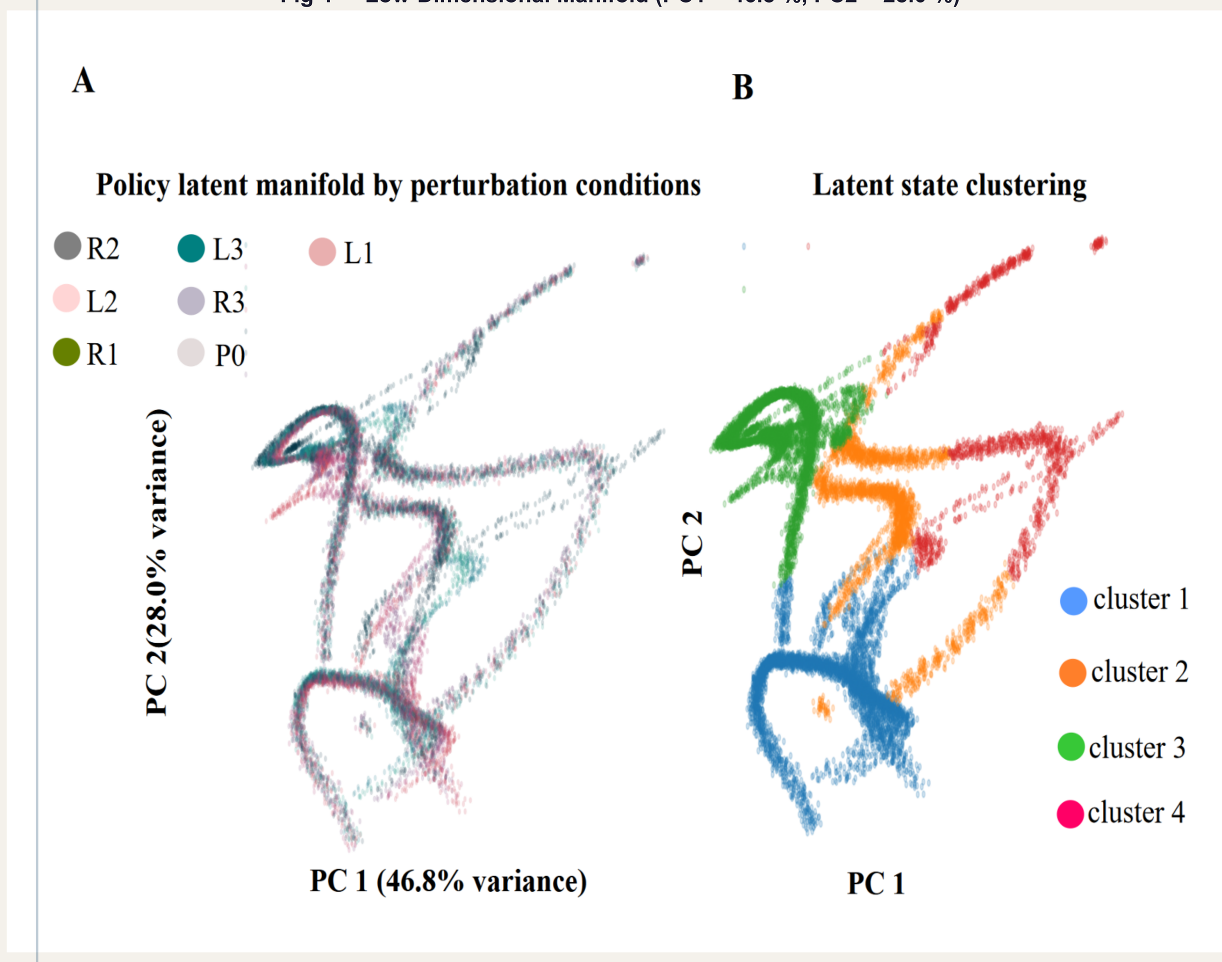
Fig 3 · Velocity Profiles: Proportional Correction, Preserved Speed



(A) Lateral velocity diverges by direction during perturbation window (t = 50–55) · (B) Total speed is invariant — adaptation is purely trajectory-level

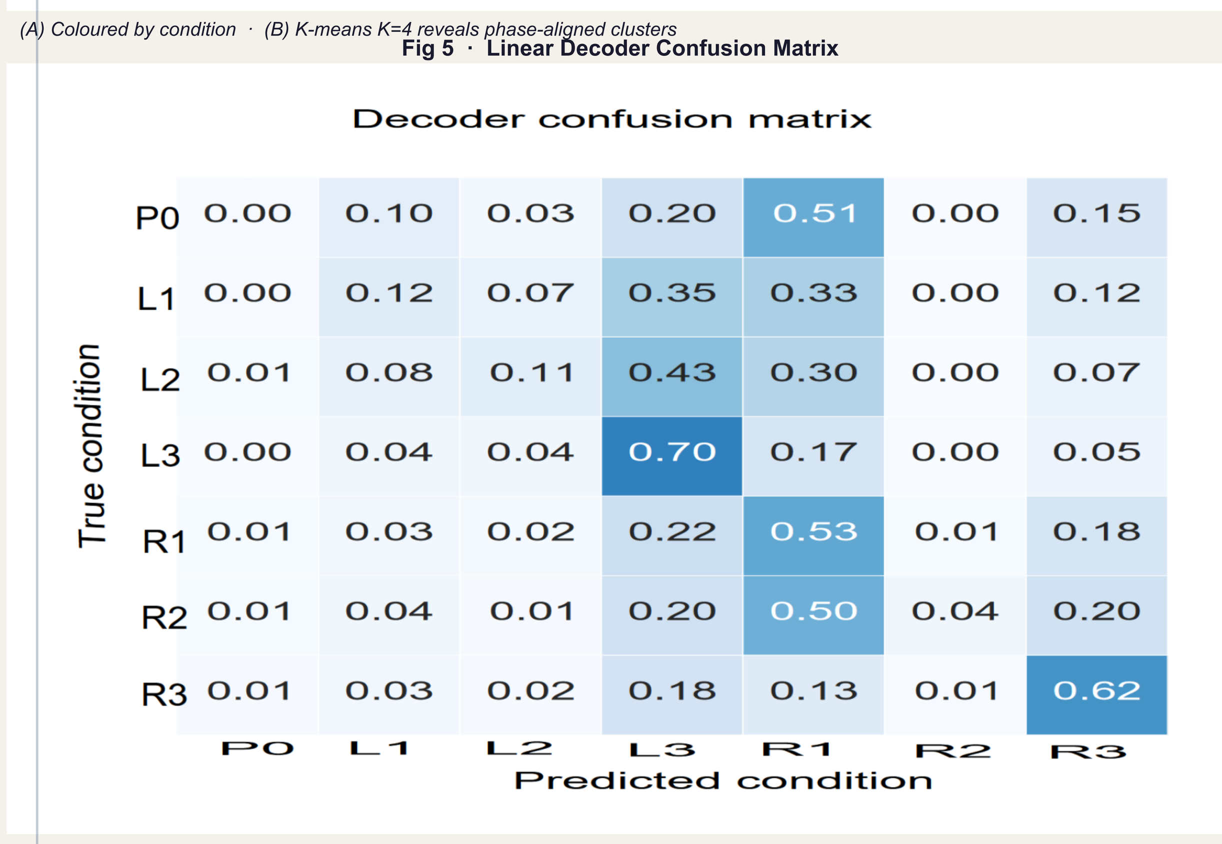
RESULTS (cont.)

Fig 4 · Low-Dimensional Manifold (PC1 = 46.8 %, PC2 = 28.0 %)



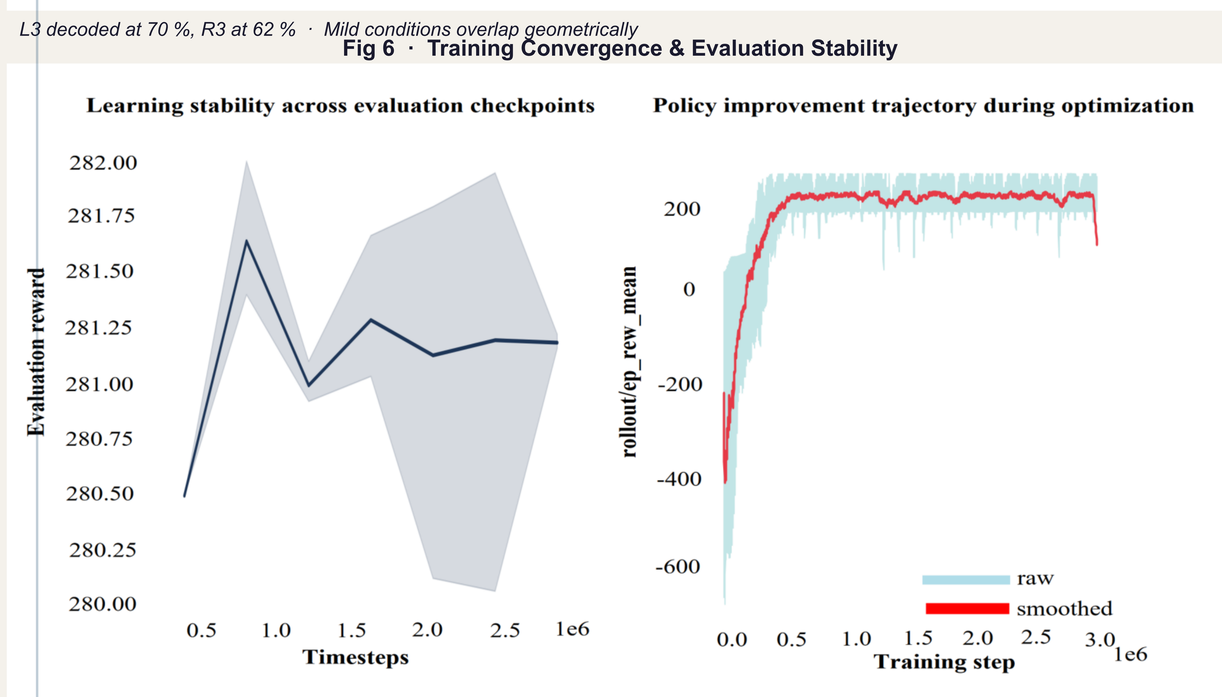
(A) Coloured by condition · (B) K-means K=4 reveals phase-aligned clusters

Fig 5 · Linear Decoder Confusion Matrix



L3 decoded at 70 %, R3 at 62 % · Mild conditions overlap geometrically

Fig 6 · Training Convergence & Evaluation Stability



Rapid convergence in first 1×10^6 steps · Stable plateau validates analyses

DISCUSSION & FUTURE WORK

- Adaptation via trajectory correction with preserved speed mirrors optimal feedback control principles in biological reaching.
- 74.8 % variance in 2 PCs matches motor cortex manifold compression documented in non-human primates and humans.
- Phase-aligned cluster structure parallels sequential neural population states identified during biological reaching.
- Direction-dependent failure asymmetry reflects learned geometry — not an inherent algorithmic limitation.

FUTURE DIRECTIONS

- Multi-seed validation to confirm representational stability.
- Recurrent architectures (LSTM) for temporal perturbation history.
- Human EMG manifold comparison in stroke rehabilitation protocol.
- Closed-loop VR platform using latent geometry as adaptive-difficulty driver.

Supported by the Connected Minds Program,
Canada First Research Excellence Fund (Grant CFREF-2022-00010)

Selected references: [1] Churchland et al., Nature 2012 · [2] Gallego et al., Neuron 2017 · [3] Sadtler et al., Nature 2014 · [4] Schulman et al., arXiv 2017 · [5] Pruszynski & Scott, Exp Brain Res 2012 · [6] Oby et al., PNAS 2019 · [7] Todorov & Jordan, Nat Neurosci 2002

Scan for code & data

Code & data: github.com/OhuePeter/NeuroRL-ObstacleAvoidance-v1.0